

xAI in Memphis: “Our lives are at stake”

By Ruby Gsanger

This past summer, as global temperatures reached record highs and protests against the presence of ICE in major cities raged around the country, a smaller, yet equally groundbreaking announcement was made: the construction of xAI’s Colossus in Memphis, TN.

xAI was founded by Elon Musk in 2023, and quickly grew to be a prominent player in the AI industry. The company purports that xAI works to “build AI specifically to advance human comprehension and capabilities.”

On June 5th, 2024, The Greater Memphis Chamber, which serves the community through “aggressively developing the region’s powerhouse economy”, announced the construction of Colossus. According to the Chamber, the world’s largest super computer would bolster the greater Memphis area. Additionally, the Chamber reported that xAI would bring 500 jobs to Memphis, a boon to the local community. Colossus is located at Frank C. Pidgeon Industrial Park in Southwest Memphis.

But this development has caused the community to take pause and beg the question: would it really impact our community positively? Not only does Musk have a sordid history as a cutthroat entrepreneur, his company has the reputation of being ethically questionable. Not only does Musk routinely withhold basic information about his AI systems, he continues to ‘build, build, build’ seemingly without considering the grave consequences of his actions.

Most notable in Memphis is the Colossus project's physical location. Colossus is being built in Southwest Memphis, historically one of the poorer zip codes in Memphis. In a 2023 census, 30.8% of the population in the area was living in poverty. The 2023 census also states that 87% of the population in South Memphis is African American. Approximately 25% (96,300 out of 394,000) of the Black population of Memphis lives in South Memphis.

In fact, roughly 8% of the community reported speaking a non-english language at home, ranging from Spanish to Yoruba. These data points bring up interesting questions about the placement of xAI's newest facility: was the decision calculated, or just one "colossus" coincidence of economic and racial discrimination?

Unfortunately, economic, racial, and environmental racism is not new to those who live in South Memphis. According to Protect Our Aquifer, a prominent sector of the community, dubbed Boxtown, fought the building of an oil pipeline, Byhalia Connection Pipeline, in 2021.

Not only would this pipeline have cut through the neighborhood and disrupted the fabric of the community, the proposed project reinforced the historic disregard for primarily Black communities. What's more, the pipeline's construction and use posed the risk of contaminating Boxtown's water sources with cancer-causing chemicals.

In the case of the pipeline, the Memphis community put up an admirable fight, and Plains All American Pipeline ultimately cancelled the project. This significant win allowed South Memphis to maintain its environmental integrity and protect its local community.

While the pipeline project was short-lived, Memphis' environmental concerns were not overcome. According to IQAir, as of October 2025, the measure of microscopic, inhalable particles in the air—PM2.5 pollution— was $8 \mu\text{g}/\text{m}$ in Memphis. In contrast, the World Health Organization cites a PM2.5 level of $5 \mu\text{g}/\text{m}$ as ideal to reduce the risk of long term breathing and vision defects. Not to mention, the Asthma and Allergy Foundation of America says the rate of allergies in Memphis is the 5th worst in the country.

Unsurprisingly, Memphians are exhibiting just as much resistance to Musk's presence in the city as they did with the Byhalia Connection Pipeline, with many local organizations fighting for South Memphis, Boxtown, and the city as a whole.

As always, there are two sides to every story. I got the chance to sit down with individuals on either side of the spectrum: the founder and operating president of Tigers Against Pollution, and an architect working at Tesla.

Orion, a college student attending the University of Memphis, founded Tigers Against Pollution after a personal brush with the effects of xAI in Memphis: "I was out in the hardwood forest, smelling the air thicken. I was like 'what's going on?'. Come to find out it's xAI in the Boxtown Area". Orion attended community meetings hosted by the Memphis Community Against Pollution (MCAP), but it wasn't enough.

The student reflects: “MCAP was doing a good job of education and data analysis. Each meeting would finish with someone asking how to help, and there really wasn’t an answer”. Orion worked hard to fill the gap of rallying and organizing the community against xAI. For her, creating Tigers Against Pollution (TAP) was the answer.

Most recently, TAP crashed xAI’s job fair at the Crosstown Concourse, and marched on xAI’s Whitehaven location. The Whitehaven plant houses 66 natural gas turbines that generate a considerable amount of energy, in turn emitting a detrimental amount of carbon emissions. TAP is working hard to advocate against xAI, having organized the Get Out of Memphis march. This event protested the presence of xAI *and* the National Guard in Memphis.

The black community has an extensive legacy of facing the effects of environmental poisoning in Memphis. Orion says, “if you look at the data, the centers all over the country are mostly in the south, in black communities. In the corporation's eyes, the black community is the path of least resistance: the members have less access to resources such as politicians and advocates”.

Communities all over the country are being steamrolled, with black communities facing the brunt of this environmental discrimination.

Data centers were proposed in other geographical areas of Memphis, such as Overton Park (roughly 50% white). In contrast to Boxtown, the plans were immediately disputed and eventually shut down, where the plan for a data center in South Memphis continued, despite community outcry.

According to Orion, xAI poses a multitude of threats to the community in Boxtown and the greater Memphis Area. “The most immediate threat is thousands and thousands of tons of toxic gases, including formaldehyde and methane, leaking into the air. These gases are three times more potent than carbon as greenhouse gases. They are literally poison.”

According to the Shelby County Health Department, while there are 27 air monitors throughout Shelby County, none are located in Boxtown. In a study conducted by Jia Chunrong and Jeffery Foran, based on data from the closest monitoring site from Boxtown, Shelby Farms (18.3 miles away), we understand that the air pollution has resulted in 4x the normal cancer rate. It’s difficult to extrapolate what it might be in Boxtown, but with greater air pollution there, there is no mistaking that it would be higher still.

Orion expands: “South Memphis communities are being hit the hardest. Air doesn’t care about boundaries. It’s going to continue to push out. Our lives are at stake. There has never been another situation like this. We have no idea what’s going to happen”.

While this diagnosis may be scary and alarming, it is the reality thousands of Memphians face. To make matters worse, the greater Memphis community is being ripped apart from the inside out. Orion claims xAI has “bought out” the University of Memphis. According to the student, U of M is pushing a focus on AI throughout all facets of the institution: research, writing, citations, and more. “They are cutting out valuable parts of learning for a University. We are there to learn”.

When a cornerstone of a community, like the University of Memphis, actively participates in its downfall, heads are bound to turn. “In Memphis, when we have a college that is being bought out by a company that is actively poisoning the families of the students of the college, it’s tragic and dangerous”.

In response to cries of outrage such as these, Musk claims that the decision to occupy South Memphis was due to availability and convenience. xAI takes no heed to the consequences the plant creates for the community. However, it might not be all bad. An architect (henceforth referred to as Arch) on the infrastructure team at Tesla’s Texas Gigafactory has an inside point of view about the center’s day-to-day operations and the positive impacts of the firm’s presence in Memphis. Due to the nature of their job title, Arch did not want to be named for the purposes of this article.

While Arch didn’t play a role in designing Colossus, they are intimately familiar with the process as part of their role at Tesla. Arch explains the process of creating a data center like xAI’s in Memphis: “The need for AI is almost exponential [in today’s digital world], so deploying these centers as quickly as possible is essential to keeping up. Chips from Nvidia, AMD, etc. are evolving extremely fast, and getting better and better. The life cycle of the chips is getting shorter and shorter, so being able to future proof for upcoming chips is essential for extending the life of a data center”.

These centers aren’t like a lot of other infrastructure: “they aren’t really buildings for people, but buildings for machines”. Companies like xAI must push forward and innovate, in order to keep

up with the ever evolving market. Choosing the location for the centers is an integral part in staying relevant and efficient.

According to Arch, the “access to utilities and infrastructure is pretty important” to the location decision for an xAI plant. Because the centers use a lot of water and power, geographical location is key. In the case of Colossus, Memphis is ideally situated as home to one of the world’s largest aquifers, holding around 100 trillion gallons of water.

Generally, local governments dictate the environmental laws and regulations companies must follow. Arch says, “Drawing power from renewable energy sources or using reclaimed and/or treated water” are a few of the factors the government has control over.

In fact, when a government allows a center such as this one to be built in its community, they are making a decision that may have a net positive impact on the surrounding area.

Arch continues: “Bigger companies like Tesla can bring tons of jobs to a local community. In Austin, for example, it’s one of the largest employers in the city. xAI is a bit different, since it doesn’t manufacture products in the same way as Tesla, but still offers many specialized engineering and computer science types of opportunities”.

Evidently, xAI boasts a unique opportunity for the Memphis community: an expanding job market, with a special set of skills. The local and state government likely had that idea in mind when approving the construction of xAI. But, the cons may still significantly outweigh the pros.

But what is there to do? Memphis is in a unique position to turn to other communities for examples. Recently, Virginia managed to flip a seat in the state legislature to democrat, by appealing to a problem that applied to the people: energy bill cost increases due to local data centers. According to the Guardian, John McAuliff, the elected official, vowed to place the burden of paying the bills on the companies that use the data, rather than the civilian community. Therefore, the local and state governments have ample ability to make change for Boxtown, South Memphis, and the entire city. All it takes is advocacy and making our voices heard.

In times like these, Memphis must come together. Governments and large corporations have a significantly harder time ignoring and resisting change when different demographics collaborate to advocate for change. When there's a close dialogue between communities, the impact of our voices is so much larger. We must take on a community first mindset, and prioritize a group of people that have been forgotten and discriminated against for far too long.

If you are interested in joining the fight against xAI in Memphis, The Memphis Community Against Pollution is attending a public hearing evaluating xAI's Methane Gas Turbine permit for December 15th at 9:30 AM . The hearing is at the University of Memphis, at the following address: University Center 499 University Drive.

There will be opportunities for individuals to voice their concerns and opinions about the permit.

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